Continuous Co-design of an Accessible CS Learning Environment with Learners who are BVI: Insights from Extended Platform Use

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Introduction

EarSketch is an **expressive learning environment** which teaches programming principles through the creation of **musical artifacts**, rendered in a digital audio workstation (DAW). The platform was **not** originally designed for **blind or visually impaired** (BVI) users, so this work engages **learners who are BVI** in **co-design workshops** to address existing accessibility barriers. Our prior workshop involved learners who had never used the platform before, providing insight into **immediate accessibility barriers** but not speaking to **longer-term** accessibility challenges or workarounds

This work is based on a follow-up co-design workshop with a **second** group of learners who are BVI and used the platform for a semester via the **Microsoft Technology Education and Learning Support** (TEALS) curriculum.

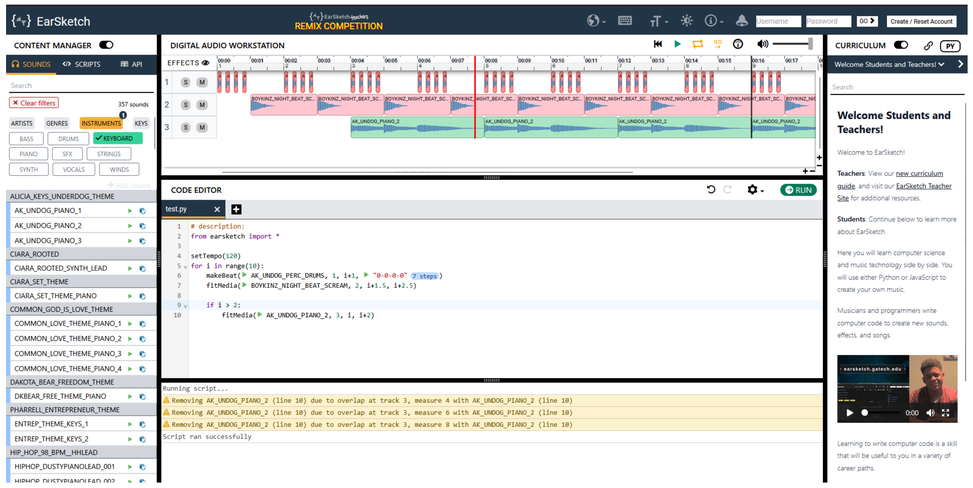
Methodology

- 5 student participants, (2 male, 2 female, 1 non-binary, aged 15-18) with various levels of visual impairment and assistive technology preference.
- 1 sighted instructor, 1 sighted TEALS aide
- 4 segments, each ~1 hour long
- Part 1: Group discussion on technology preferences and EarSketch barriers encountered during semester
- Part 2: Speculative prototyping activity
- Part 3: AI roleplay activity
- Part 4: Students evaluate new features in EarSketch

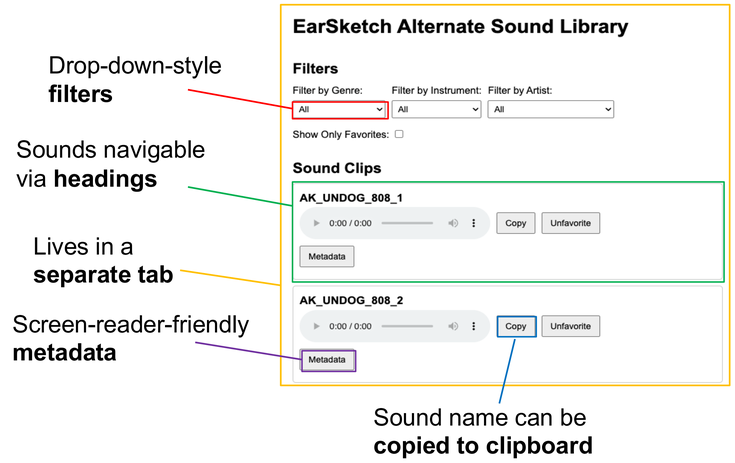
EarSketch Website

JAWS Walkthrough

The EarSketch Interface



Alternate Sound Browser Created by TEALS Aide



Discussion

The desires for **hierarchy**, **jump points**, **traceability** to a prior location, and detached, **modular views** are all connected to a lack of **context** due to linearly-constricted navigation methods. They are potential ways to **orient** within the interface and maintain context across its parts rather than re-traversing linearly. Additionally, screen reader output needs to be **unobtrusive** but also **contextually meaningful**.

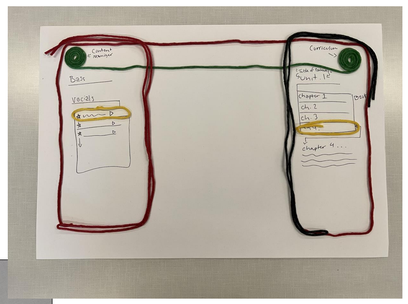
Multi-paned interfaces should provide **structural orientation**, such as **hierarchical landmarks**, to center navigation *on* the panes rather than *around* them.

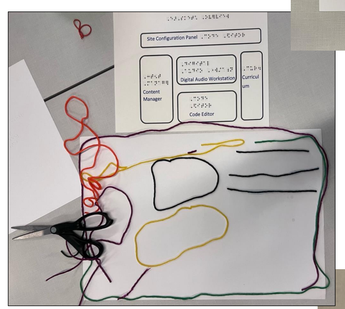
Acknowledgements

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Wikki Stix Prototypes Designed by Learners

- Highlights **information** panes, I.E. **Content Manager** and **Curriculum**





- **Horizontal** lines representing **stacked** curriculum content
- Panes differentiated with **borders**

- Main navigation **loop** between **Code Editor** and **DAW**
- Tactile graphic serves as base to depict **workflow**

